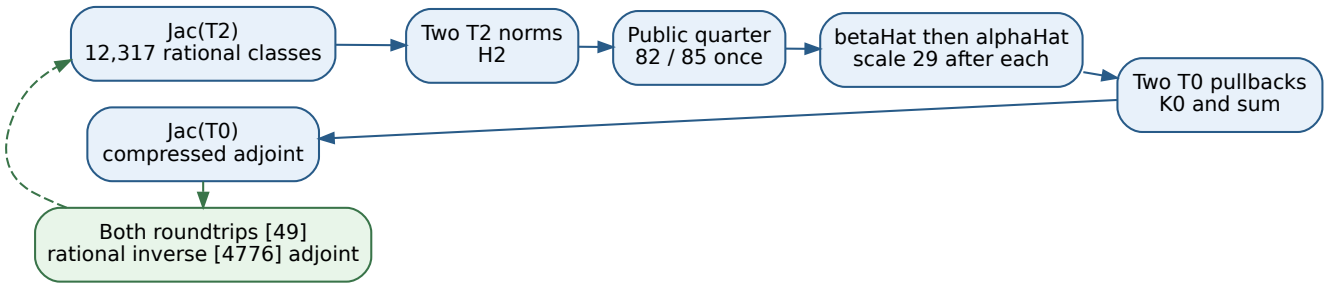
Compressed reverse evaluation and rational recovery

2026-11-12 · fixed F101 chain · scientific standing unsigned

Isolated native01/PCC4ac90f accepts the pure second adjoint and compressed two-hop adjoint on all12,317 classes. Both cases compare explicit old outputs, check bijection,negation,fixed-partner sums,identity,model and wrong-quarter controls. Both whole-domain norm7/norm49 compositions pass. Every109/113 original source reverse196 and public recovery104/64 passes. Accepted pure first adjoint dcbb58 and global/Picard inputs are explicit dependencies; this is local correctness evidence,not an independent kernel certificate. [pcc:unverified]



The reverse formula

Taking polarized adjoints of the accepted global4F=K2 beta alpha H0 gives 4FHat=K0 alphaHat betaHat H2. Public quarters82/85 evaluate this identity on the odd rational groups. Only outer T2 norms and T0 pullbacks occur; there is no intermediate T1 Jacobian class. Geometric degree stays2401. REVERSE_CHAIN_PROOF.md/001dff explains the normalization and dependencies. [pcc:unverified]

Reverse map	Kernel cubic,low first	Actual short model
betaHat109	[13,62,81,1]	(59,27) to (26,58)
betaHat113	[11,64,23,1]	(23,7) to (69,50)
alphaHat109	[58,26,58,1]	(26,58) to (1,46)
alphaHat113	[29,88,45,1]	(69,50) to (44,21)

Each normalized Kohel dual needs checked scale29=1/7 before the next stage. Old norm identities give FHat F=F FHat=[49]. Since49 times4776 equals 1+19 times12317,the rational inverse is [4776]FHat. This scalar is distinct from embedded-source recovery:the final H0K0 contributes4,so source reversal is[196] with inverses104mod109 and64mod113. [pcc:unverified]

Faithful formal consequences

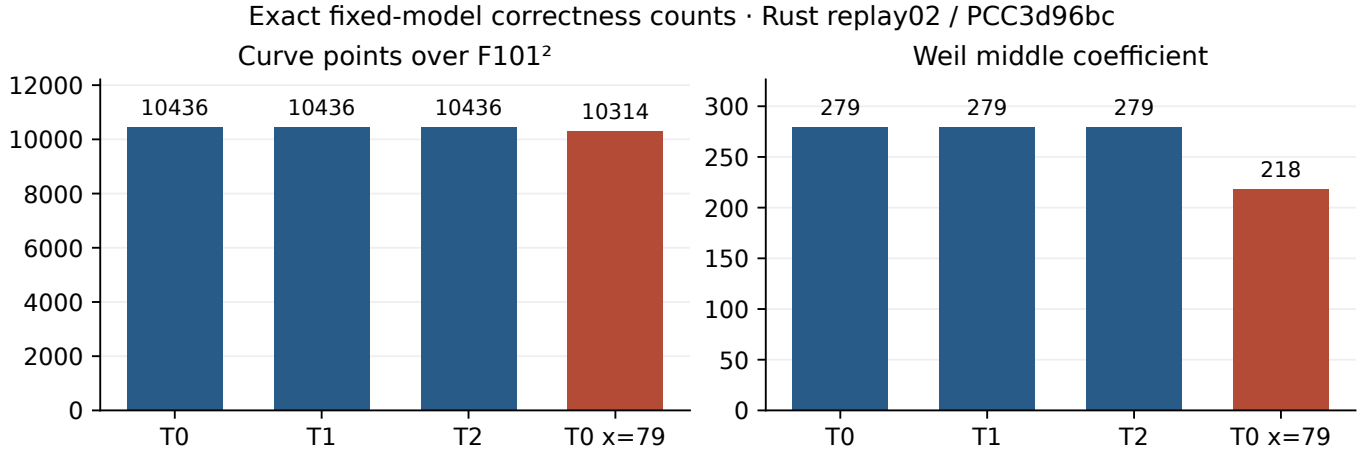
ReverseRecovery.recovery49_order12317/082015 and two_sided_inverse_from_norm49/71f035 are individually bound,locally kernel checked module statements. The second assumes both norm49 identities, forward integer-linearity and annihilation by12317 on both groups;it proves both inverse identities. Initial carrier-elaboration failuredb6ba8 is retained. Actual varieties,Picard maps,orders and norm realizations remain separate written/native dependencies;scientific signing is separate. [pcc:unverified]

Actual integration checkpoint

Ready cryptanalysis1507 at93587b9 has the full first-adjoint check25native/ 92reference unit plus five integrations/one historical ignored d744cf, writteneceeb1/report8248ae/packd2f0e5. Its own focusedCI was dispatched09:23, next check10:13. Parent1506 exact43180569278 succeeds/full64e1d9. New actual second/compressed branch is running the complete98-test reference check on 13frozen source02 files,d0ad0a;no completion is asserted here. [pcc:unverified]

Spectra,negative controls and proof boundaries

Independent Rust3d96bc checks the actual T0/T1/T2 sextics using existing Zp/FpR/poly engines and different quadratic basis z^2-3. Every101/10,201 Euler character agrees with a full square-membership table. All four curves are smooth with two rational infinity points. No producer map is imported. External CT39 at a30c63d1 is credited;its executables were not run here. [pcc:unverified]



The three actual curves each give N1=120,N2=10436. Newton identities yield $P(T)=T^4+18T^3+279T^2+1818T+10201=(T^2+7T+101)(T^2+11T+101)$,$P(1)=12317$. The smooth T0 x-coefficient75→79 mutation keeps N1=120 but gives N2=10314, middle218 and order12256. Thus one curve count cannot identify the genus2 isogeny class. Galbraith Chapter10.7/Newton/Tate supplies the written spectral interpretation;it supplies no selected correspondence or kernel. [pcc:unverified]

All eligible torsion depths,with explicit exceptions

For $f=T^2+7T+101$,$g=T^2+11T+101$,put $A=-(T+7)g$,$B=(T+11)f$. Then $A+B=404$ and $AB=-(T+7)(T+11)fg$. At every prime ell except2,101 and depthn,the inverse of404 modulo ell to powern makes A(pi),B(pi) complementary idempotents on the corresponding torsion. They preserve Frobenius-stable subgroups and commute with F101 homomorphisms. At2 the factors coincide;at101 they share zero and etaleness need not hold. No integral global product follows. [pcc:unverified]

SpectralControls.c783fc/032f9e/d261e9 formalize the fixed ring implications, unique additive-projector decomposition and cubic-polynomial naturality. Older OddKernelProjectors already supplies generic CRT algebra. The new statements do not construct actual Tate modules or subgroup schemes. SPECTRAL_CONTROL_PROOF.md/76d876 gives the written bridges. [pcc:unverified]

Rational and tangent agreement can hide a different kernel

CT38′s m1=1244018 and m0=4976069 act as1 on every12317-annihilated point and every101-annihilated tangent vector. On7-annihilated points they act as−1 and0. SpectralControls.rational_tangent_shadow/28f4e8 proves that precise module statement. Multiplication degrees and geometric7-torsion realization remain written Milne inputs. Independent global degree49/2401 and degree-bounded quartic identification reject these high-degree impostors; rational tests remain a separate obligation. [pcc:unverified]

Changed standalone21/strictClippy/fmt/sourceguards3d96bc passes. Six new individually bound Lean lemmas give144written declarations/28modules;all scientific standing remains unsigned. Prior root9084 exact43185969853 succeeds25native/20standalone/138written/26modules,full411a94;the new scope requires its own exact-headCI. Earlier manual-divceil9eaf22,carrierdb6ba8 and prose-marker failures remain. Old outer.svg/geometry.svg were checked: their forward/geometric scopes stay accurate;new reverse/spectral figures cover these additions. Live-budget admission9e1812 blocks paid cost runs. [pcc:unverified]